Project 1: Design for Deflection

Phase 1: Castigliano's Method and FEA Practice and Report

Ryan Beyer, Aidan Hoidal-Bui, Anne Jumper

Introduction:

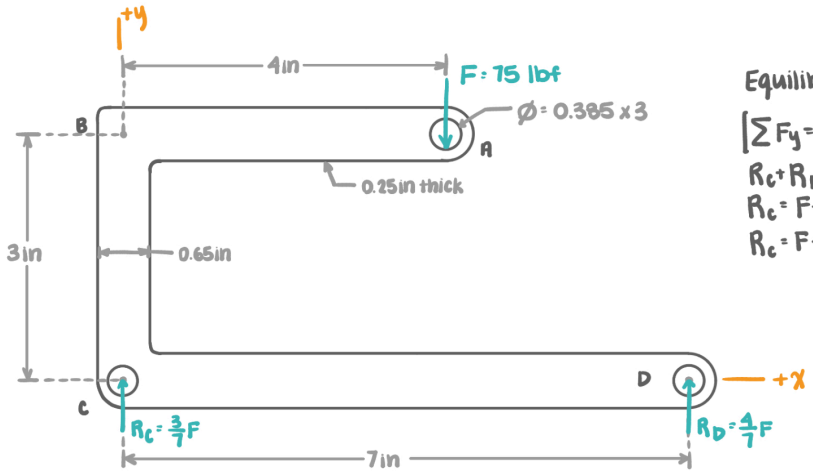
In completing Phase I of this Design for Deflection project, we began by analyzing the two structures (prismatic and tapered) using Castigliano's Theorem with respect to the 75 lbf force applied at the cantilevered top right points (point A) of the structures. We broke each structure into smaller sections to ease the analysis process: the top cantilevered beam AB under bending, the vertical beam BC under compression and bending, and the horizontal simply supported beam CD under bending. We chose to neglect shear bending for our analysis given that the beams are slender and the large applied force causes deflection mostly due to bending. To calculate deflection, we drew free body diagrams for the entirety of each structure and for the three sections and then applied Castigliano's theorem to each section for each mode of deformation. By taking the partial derivative (with respect to the applied force) of the strain energy for each mode of deformation, we were able to compute the resulting deflection in each section. Summing these deflections returned the overall deflection of the structure at point A.

We then used FEA simulation to replicate loading conditions and analyze the resulting deflection at the cantilevered point A. This process began by assigning the fixtures to the structures on the cylindrical faces of the bottom pins. At the right pin, a radial fixture was used to allow the structure to rotate around the pin and an axial fixture was used to prevent motion forward and backward along the pin's axis. The left pin utilizes a strut to allow for lateral movement so that the bottom beam can bend. A fixture is applied to the bottom of this strut to achieve this. Finally, the 75 lbf external load is applied. To run the finite element analysis, a mesh is applied to the structure which is coarser along the lengths of the beams to optimize computing time and finer at the corners and pins where higher resolution is desirable to most accurately observe the deflection. Running the simulations for both structures returned colored deflection contour plots. Additional plots showing the deflection probed along the split-line of the cantilevered pin were also obtained.

Analysis:

<u>Prismatic Beam</u>

Free Body Diagram



Equilibrium Analysis:

$[\Sigma F_y = 0]$

$R_C + R_D - F = 0$

$R_C = F - R_D$

$R_C = F - \frac{4}{7}F$

$[\Sigma M_C = 0]$

$(7in)R_D - (4in)F = 0$

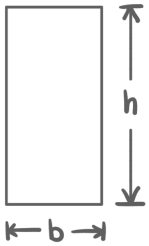
$\boxed{R_D = \frac{4}{7}F}$

$\boxed{R_C = \frac{3}{7}F}$

Castigliano's Analysis

Calculating Second Moment of Inertia

Cross section of beam:



$b = 0.25\,in$

$h = 0.65\,in$

$A = bh = (0.25in)(0.65in) = 0.1625\,in^2$

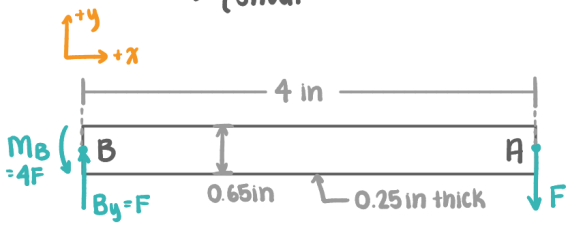
$I_x = \frac{bh^3}{12} = \frac{(0.25in)(0.65in)^3}{12} \Rightarrow \boxed{I = 0.00572\,in^4}$

$\boxed{E = 10 \times 10^6\,psi}$

Break structure into 3 rectangular beams for simplified analysis:

Beam AB:

Under bending {normal ~~shear~~



Deflection due to Bending



$M(x) = -Fx$

$\frac{\partial M}{\partial F} = -x$

$\delta_{AB} = \frac{\partial U}{\partial F} = \int \frac{1}{EI}\left(M\frac{\partial M}{\partial F}\right)dx$

$= \frac{1}{EI}\int_0^4 (-Fx)(-x)\,dx$

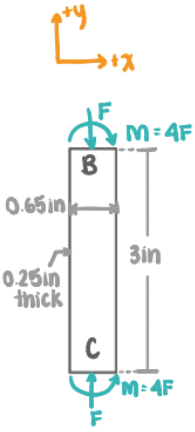
$= \frac{1}{(10\times10^6\frac{lb}{in^2})(0.00572\,in^4)}\left[\frac{Fx^3}{3}\right]\Big|_0^4$

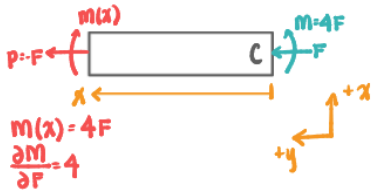
$= \frac{1}{(10\times10^6\frac{lb}{in^2})(0.00572\,in^4)}\frac{(75lb)(4in)^3}{3}$

$\boxed{\delta_{AB} = 0.028\,in}$

Beam BC:

Under compression and bending {normal ~~shear~~





Deflection due to Bending

$\delta_{BC,b} = \frac{\partial U}{\partial F} = \int \frac{1}{EI}\left(M\frac{\partial M}{\partial F}\right)dx$

$= \frac{1}{EI}\int_0^3 16F\,dx$

$= \frac{1}{(10\times10^6\frac{lb}{in^2})(0.0057in^4)}\left[16Fx\right]_0^3$

$= \frac{1}{(10\times10^6\frac{lb}{in^2})(0.0057in^4)}(16(75lb)(3in))$

$\boxed{\delta_{BC,b} = 0.063\,in}$

Deflection due to Compression

$\delta_{BC,c} = \frac{\partial U}{\partial F} = \int \frac{1}{AE}\left(P\frac{\partial P}{\partial F}\right)dx$

$= \frac{1}{AE}\int_0^3 F\,dx$

$= \frac{1}{(0.1625in^2)(10\times10^6\frac{lb}{in^2})}\left[Fx\right]_0^3$

$= \frac{(75lb)(3in)}{(0.1625in^2)(10\times10^6\frac{lb}{in^2})}$

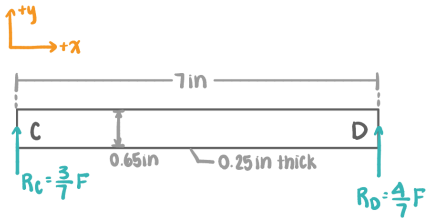
$\boxed{\delta_{BC,c} = 0.000138\,in}$

Total Deflection:

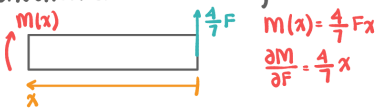
$\delta_{BC} = \delta_{BC,b} + \delta_{BC,c} = 0.063in\ \ 0.000138in$

$\boxed{\delta_{BC} = 0.0634\,in}$

Beam CD:
Under Bending {normal / ~~shear~~



Deflection due to Bending



$M(x) = \frac{4}{7}Fx$

$\frac{\partial M}{\partial F} = \frac{4}{7}x$

$$\delta_{CD} = \frac{\partial U}{\partial F} = \frac{1}{EI}\int M\left(\frac{\partial M}{\partial F}\right)dx$$

$$= \frac{1}{EI}\int_0^7 \frac{4}{7}Fx\left(\frac{4}{7}x\right)dx$$

$$= \frac{1}{(10\times10^6 \frac{lb}{in^2})(0.00572\,in^4)}\left[\frac{16Fx^3}{49\cdot3}\right]\Big|_0^7$$

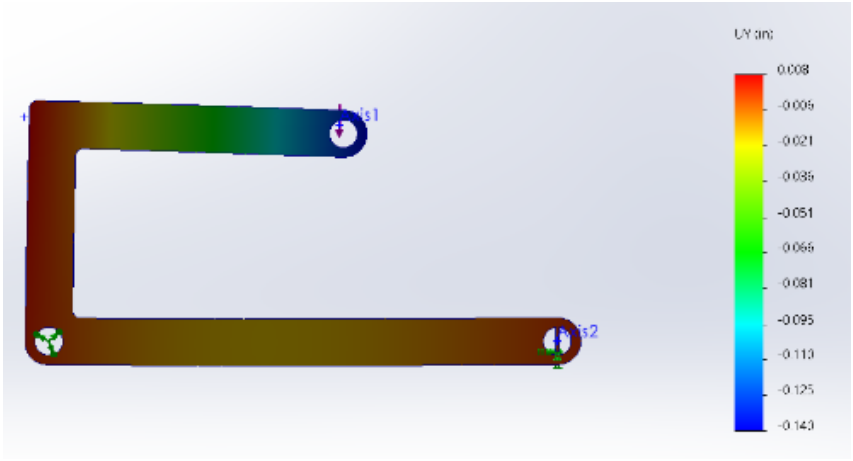
$$= \frac{1}{(10\times10^6 \frac{lb}{in^2})(0.00572\,in^4)}\frac{16(75lb)(7in)^3}{49\cdot3}$$

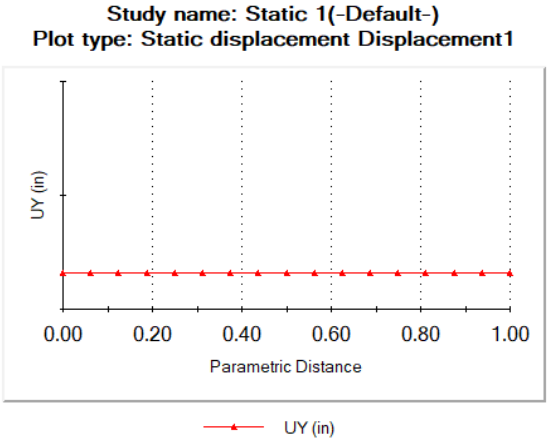
$\delta_{CD} = 0.0489\,in$

Total Deflection: $\delta_{total} = \delta_{AB} + \delta_{BC} + \delta_{CD} = 0.028 + 0.0634 + 0.0489\,in$

$\delta_{total} = 0.1403\,in$

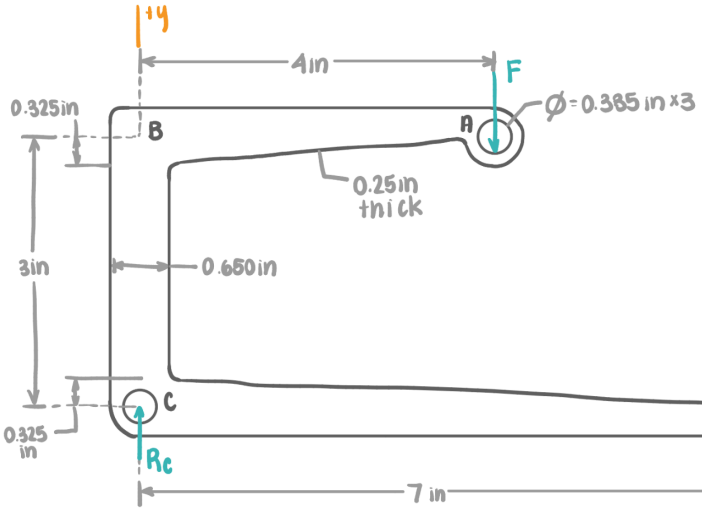
FEA Analysis





Average deflection at Point A = 0.128 in

Tapered Beam



Equilibrium Analysis:

$[\sum F_y = 0]$

$R_C + R_D - F = 0$

$R_C = F - R_D$

$R_C = F - \frac{4}{7}F$

$[\sum M_C = 0]$

$(7in)R_D - (4in)F = 0$

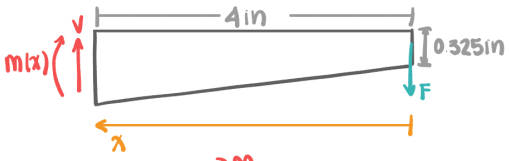
$R_D = \frac{4}{7}F$

$R_C = \frac{3}{7}F$

Break structure into 3 rectangular beams for simplified analysis:

Beam AB:

Under bending {normal, ~~shear~~}



$M(x) = -Fx \quad \frac{\partial M}{\partial F} = -x$

$I(x) = \frac{bh^3}{12}$ } $h(x) = 0.325 + \frac{0.325}{4}x$ in, $b = 0.25$in

$$I(x) = \frac{0.25\left(0.325 + \frac{0.325}{4}x\right)^3}{12}$$

Deflection due to Bending

$$\delta_{AB} = \int \frac{1}{EI} M(x) \frac{\partial M}{\partial F} dx$$

$$= \frac{1}{E}\int \frac{M(x)}{I(x)} \frac{\partial M}{\partial F} dx$$

$$= \frac{1}{(10\times10^6)} \int_0^4 \frac{Fx^2}{I(x)} dx$$

Solved using Symbolab integral calculator

$\delta_{AB} = 0.0457$in

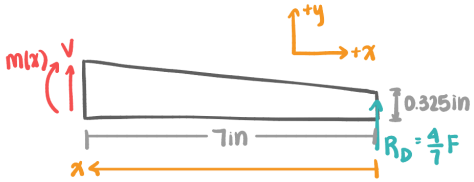
Beam BC:

Equivalent size and loading conditions to prismatic case

$\delta_{BC} = 0.0634$in

Beam CD:

Under Bending {normal, ~~shear~~}



$M(x) = \frac{4}{7}Fx \quad \frac{\partial M}{\partial F} = \frac{4}{7}x$

$I(x) = \frac{bh^3}{12}$ } $h(x) = 0.325 + \frac{0.325}{7}x$ in, $b = 0.25$in

$$I(x) = \frac{0.25\left(0.325 + \frac{0.325}{7}x\right)^3}{12}$$

Deflection due to bending

$$\delta_{CD} = \int \frac{1}{EI(x)} M(x) \frac{\partial M}{\partial F} dx$$

$$= \frac{1}{10\times10^6 \frac{lb}{in^2}} \int_0^7 \frac{16Fx^2}{49\left(\frac{0.25\left(0.325+\frac{0.325}{7}x\right)}{12}\right)} dx$$

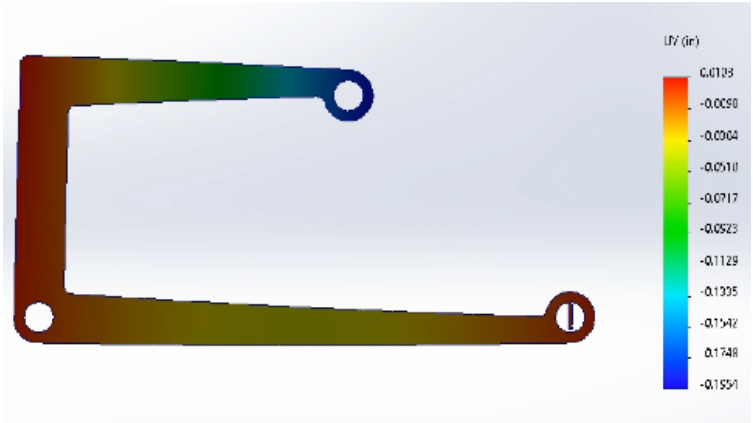
$\delta_{CD} = 0.08$in

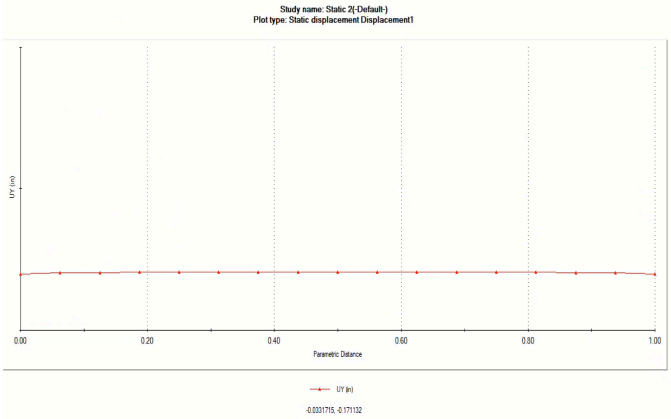
Solved using Symbolab integral calculator

Total Deflection: $\delta_{total} = \delta_{AB} + \delta_{BC} + \delta_{CD} = 0.0457 + 0.0634 + 0.08$ in

$\delta_{total} = 0.1891$ in

FEA Analysis





Average Deflection at Point A = 0.178 in

Discussion:

Using Castigliano's Theorem, we solved for the deflection of the two structures (prismatic and tapered). We split the structure up into three beams to simplify the geometry, and then solved for the deflection caused by bending in all three beams and compression in the second beam. The method used was the same for both structures, however, the area moment of inertia is a function of x in the tapered structure. To solve the integrals for the tapered structure, we used the SymboLab integral calculator. The total deflection in the prismatic structure was found to be -0.1403 in., and the deflection in the tapered structure was found to be -0.1891 in.

Using SolidWorks Simulation, we performed FEA to determine and visualize the vertical deflection of each beam under the applied load. The simulation predicted the deflection to be -0.128 in for the prismatic structure and -0.178 in for the tapered structure. These results were in strong agreement with those found by our analytical calculations using Castigliano's Theorem.

The percentage errors between the Castigliano's Theorem values and the SolidWorks FEA values for the prismatic structure and tapered structure were 9.61% and 6.23%, respectively, confirming the accuracy of both our modeling and theoretical methods. Our results from both methods indicate that the tapered structure experiences greater overall deflection than the prismatic structure. This is expected, as the prismatic beam maintains a constant cross-sectional area along its length, providing greater stiffness and resistance to bending compared to the tapered beam, whose reduced cross-sectional area near the free end allows for increased deformation under load.